

Name: _____ Neptun code: _____

(KTXFI2EBNF) Physics I. Signature Retake Exam

May 26, 2026

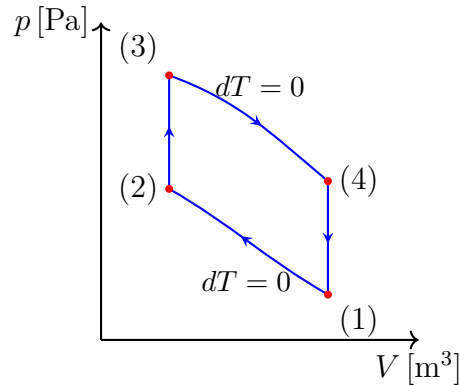
I. Mechanics

1. A projectile is launched at an angle of 60° with an initial velocity of 100 m/s. Calculate:
 - (a) The time to reach the peak of the trajectory.
 - (b) The maximum height.
 - (c) The horizontal range (impact distance).
 - (d) The total time of flight. $(g = 9.81 \text{ m/s}^2)$
2. At what distance from the Uranus does an geosynchronous satellite orbit in the plane of the Equator if it remains constantly above the same point on the Uranus's surface? ($M_{Uranus} = 8.68 \cdot 10^{25} \text{ kg}$, $T_{Uranus} = -0,72 \text{ day}$, $R_{Uranus} = 25300 \text{ km}$, $G = 6.6743 \cdot 10^{-11} \text{ Nm}^2/\text{kg}^2$.)
3. A 5 kg object slides down from the top of a 2-meter-high inclined plane that makes an angle of 15° with the horizontal. What is the velocity of the object when it reaches the bottom of the incline, starting from rest at the top, if:
 - (a) there is no friction between the object and the incline?
 - (b) the coefficient of friction is 0.2? $(g = 9.81 \text{ m/s}^2)$
4. A string is wound around a solid cylinder with radius $R = 0.6 \text{ m}$ and mass $M = 50 \text{ kg}$. Two objects with a combined mass of 25 kg are hanging from the ends of the string. Upon release, the cylinder accelerates with an angular acceleration of $\beta = 5 \text{ rad/s}^2$. The string does not slip. $(g = 9.81 \text{ m/s}^2)$
 - (a) Find the individual masses (m_1, m_2) of the two objects.
 - (b) Calculate the total kinetic energy of the system at $t = 5 \text{ s}$.
5. Estimate the average depth of the Atlantic Ocean if a tsunami takes four hours to travel from Reykjavík to Lisbon. (Note: The propagation speed of shallow-water waves is $v = \sqrt{gh}$, where $g = 9.81 \text{ m/s}^2$. The distance is 3000 km.)

II. Thermodynamics

6. We place a copper piece with a mass of 500 gramms and a temperature of 40°C into 1250 gramms of water at 70°C . What will be the common temperature? The specific heat capacity of copper is $3.85 \cdot 10^2 \frac{\text{J}}{\text{kg K}}$ and the specific heat capacity of water is $4.19 \cdot 10^3 \frac{\text{J}}{\text{kg K}}$.
7. At a temperature of 50°C , a steel shaft with a diameter of 11.21 cm needs to be fitted with an aluminum ring with an inner diameter of 11.18 cm at the same temperature. The linear thermal expansion coefficient of steel is $12 \cdot 10^{-6} \frac{1}{\text{K}}$, and the linear thermal expansion coefficient of aluminum is $28.7 \cdot 10^{-6} \frac{1}{\text{K}}$. What temperature should the ring be heated to?

8. The atmospheric balloon floating at an altitude of several kilometers has a volume of 1800 m^3 . The temperature and pressure of the helium inside it are the same as the surrounding -40°C and $5 \cdot 10^3 \text{ Pa}$, respectively.
- How many moles of helium are in the balloon?
 - What is the mass of the helium gas? ($M = 4 \frac{\text{g}}{\text{mol}}$)
 - What was the volume of the balloon at the time of launch on Earth, if the gas inside the balloon was characterized by 0°C and 10^5 Pa pressure?
- ($R = 8.31 \frac{\text{J}}{\text{mol}\cdot\text{K}}$)
9. The ideal cycle of an externally heated heat engine, shown in the diagram, differs from the Otto engine cycle in that it contains isothermal processes instead of adiabatic ones.

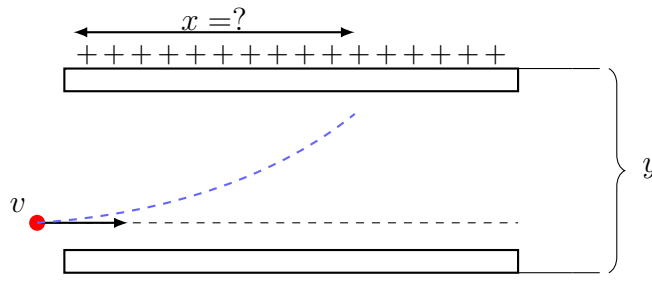


What is the efficiency of the cycle if $p_3 = 2p_2$, the compression ratio is 8, and the cycle is performed with a two-atomic (diatomic) gas? What is the efficiency of the Carnot cycle taking place between the same temperature limits? ($R = 8.31 \frac{\text{J}}{\text{mol}\cdot\text{K}}$)

10. What is the entropy change if the volume of $m = 2 \text{ g}$ of hydrogen changes from $V_1 = 30 \text{ l}$ to $V_2 = 120 \text{ l}$, and its temperature changes from $t_1 = 80^\circ\text{C}$ to $t_2 = 400^\circ\text{C}$? The molar mass is $M = 2 \text{ g/mol}$, $R = 8.31 \frac{\text{J}}{\text{mol}\cdot\text{K}}$.

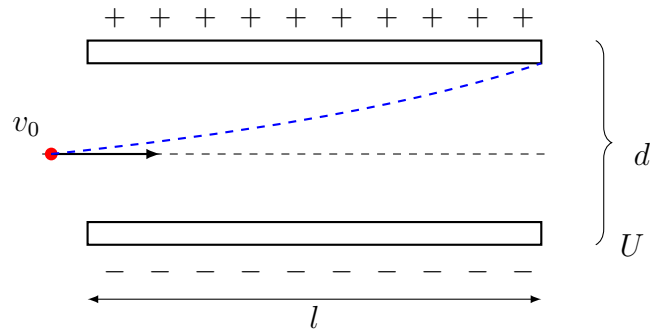
III. Electrostatics, Waves, and Optics

11. Determine the acceleration of a proton $m = 1.67 \cdot 10^{-27} \text{ kg}$ in an electric field of strength $E = 300 \text{ N/C}$. How many times greater is this acceleration than the acceleration due to gravity? ($e = 1.6 \cdot 10^{-19} \text{ C}$, $g = 9.81 \text{ m/s}^2$)
12. An electron is launched with an initial velocity of $v_0 = 3 \cdot 10^6 \text{ m/s}$ between two parallel charged plates, as shown in the figure. If the electric field strength between the plates is $E = 3 \text{ kN/C}$, where does the electron hit the upper plate? Assume that the arrangement is in vacuum.



($m_e = 9.1 \cdot 10^{-31} \text{ kg}$, $e = 1.6 \cdot 10^{-19} \text{ C}$)

13. An electron enters the region between the plate pair shown in the figure halfway between the plates. The length of the plates is $l = 20\text{ cm}$, their separation is $d = 2\text{ cm}$, and the voltage between them is $U = 500\text{ V}$. What must the initial velocity of the electron be if we want it to just barely leave the region between the plates? ($m_e = 9.1 \cdot 10^{-31}\text{ kg}$, $e = 1.6 \cdot 10^{-19}\text{ C}$)



Grades: Rejected, Teacher's signature.

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